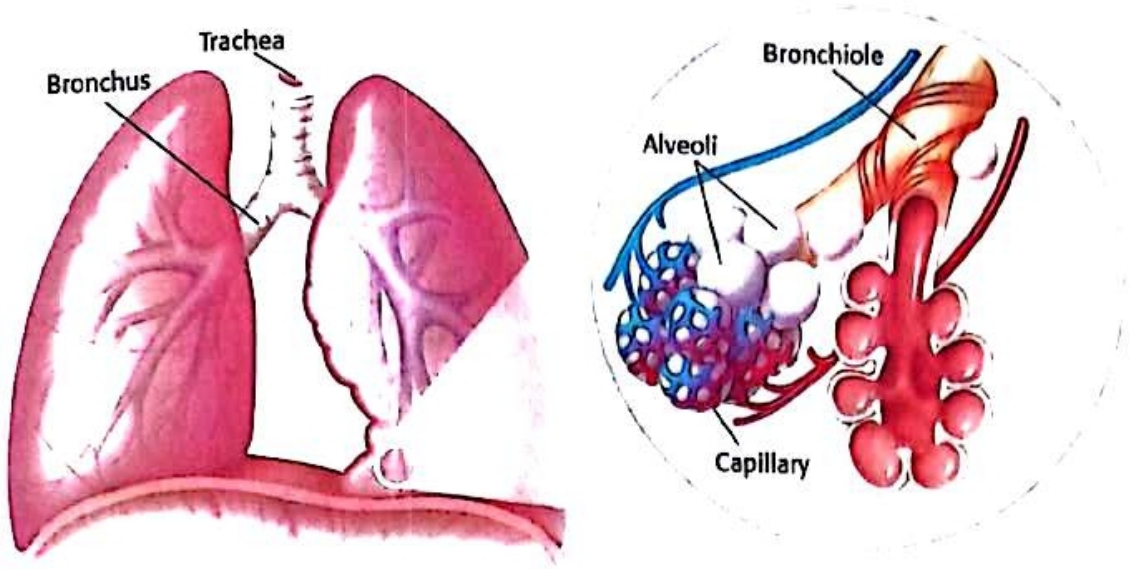


Figure 2 *Inside your lungs, the bronchi branch into bronchioles. The bronchioles lead to tiny sacs called alveoli.*

Nose, Pharynx, and Larynx

Your *nose* is the main passageway into and out of the respiratory system. Air can be breathed in through and out of the nose. Air can also enter and leave through the mouth.

From the nose, air flows into the **pharynx** (FAR ingks), or throat. Food and drink also travel through the pharynx on the way to the stomach. The pharynx branches into two tubes. One tube, the *esophagus*, leads to the stomach. The other tube is the larynx (LAR ingks). The larynx leads to the lungs.

The **larynx** is the part of the throat that contains the vocal cords. The *vocal cords* are a pair of elastic bands that stretch across the larynx. Muscles connected to the larynx control how much the vocal cords are stretched. When air flows between the vocal cords, the cords vibrate. These vibrations make sound.

Trachea

The larynx guards the entrance to a large tube called the **trachea** (TRAY kee uh), or windpipe. Your body has two large, spongelike lungs. The trachea, shown in **Figure 2**, is the passageway for air traveling from the larynx to the lungs.

Bronchi and Alveoli

The trachea splits into two branches called **bronchi** (BRAHNG KIE) (singular, *bronchus*). One bronchus connects to each lung. Each bronchus branches into smaller tubes that are called *bronchioles* (BRAHNG kee OHLZ). In the lungs, each bronchiole branches to form tiny sacs that are called **alveoli** (al VEE uh LIE) (singular, *alveolus*).

Reading Check **Describe the flow of air from your nose to your alveoli.** (*See the Appendix for answers to Reading Checks.*)

pharynx the passage from the mouth to the larynx and esophagus

larynx the area of the throat that contains the vocal cords and produces vocal sounds

trachea the tube that connects the larynx to the lungs

bronchus one of the two tubes that connect the lungs with the trachea

alveoli any of the tiny air sacs of the lungs where oxygen and carbon dioxide are exchanged

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SECTION 4

The Respiratory System

Breathing—you do it all the time. You're doing it right now. You hardly ever think about it, though, unless you suddenly can't breathe.

Then, it becomes very clear that you have to breathe in order to live. But why is breathing important? Your body needs oxygen in order to get energy from the foods you eat. Breathing makes this process possible.

What You Will Learn

- Describe the parts of the respiratory system and their functions.
- Explain how breathing happens.
- Discuss the relationship between the respiratory system and the cardiovascular system.
- Identify two respiratory disorders.

Vocabulary

respiration
respiratory system
pharynx
larynx
trachea
bronchus
alveoli

READING STRATEGY

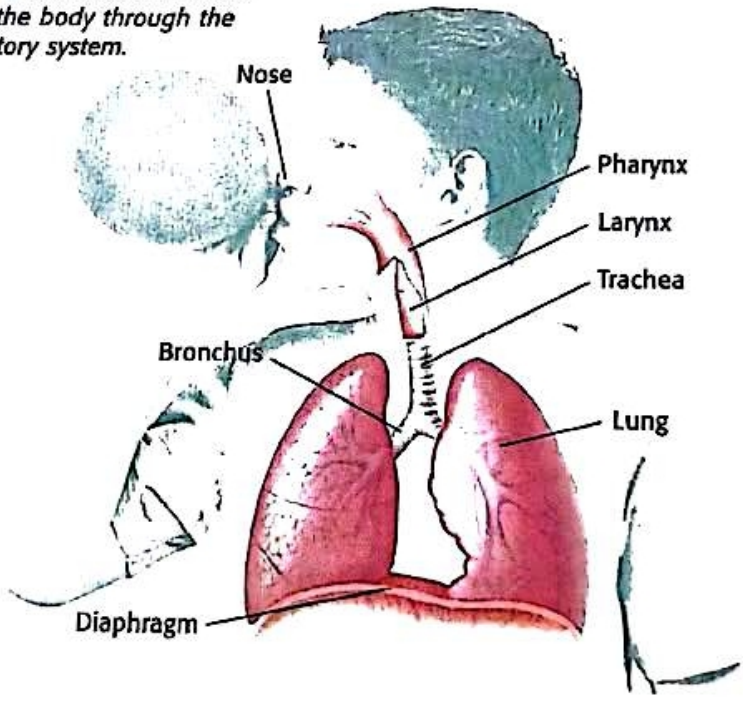
Reading Organizer As you read this section, make a flowchart of the steps of the process of respiration.

Respiration and the Respiratory System

The words *breathing* and *respiration* are often used to mean the same thing. However, breathing is only one part of respiration. **Respiration** is the process by which a body gets and uses oxygen and releases carbon dioxide and water. Respiration is divided into two parts. The first part is breathing, which involves inhaling and exhaling. The second part is cellular respiration, which involves chemical reactions that release energy from food.

Breathing is made possible by your respiratory system. The **respiratory system** is the group of organs that take in oxygen and get rid of carbon dioxide. The nose, throat, lungs, and passageways that lead to the lungs make up the respiratory system. **Figure 1** shows the parts of the respiratory system.

Figure 1 *Air moves into and out of the body through the respiratory system.*



respiration the exchange of oxygen and carbon dioxide between living cells and their environment; includes breathing and cellular respiration

respiratory system a collection of organs whose primary function is to take in oxygen and expel carbon dioxide

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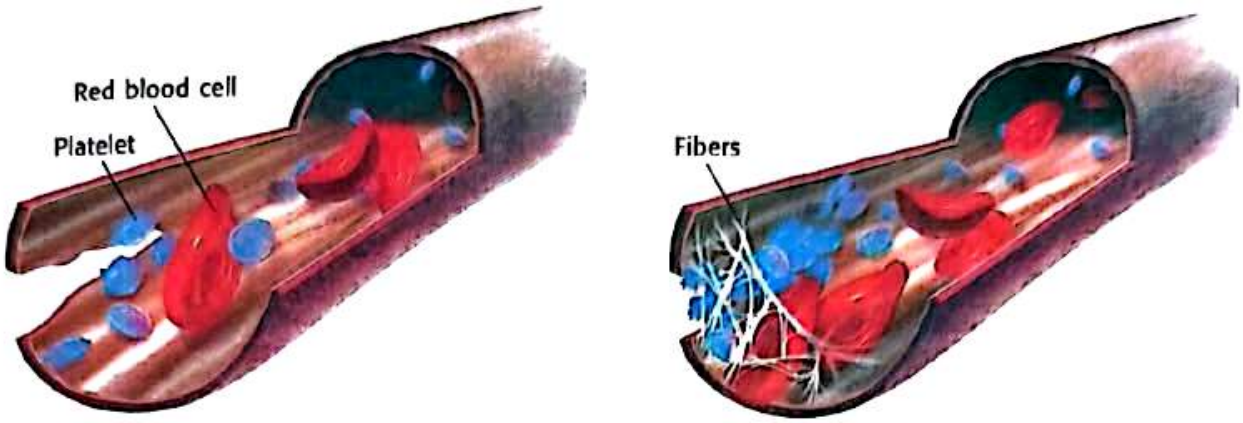


Figure 2 *Platelets release chemicals in damaged vessels and cause fibers to form. The fibers make a "net" that traps blood cells and stops bleeding.*

Platelets

Drifting among the blood cells are tiny particles called platelets. *Platelets* are pieces of larger cells found in bone marrow. These larger cells remain in the bone marrow, but fragments are pinched off and enter the bloodstream as platelets. Platelets last for only 5 to 10 days, but they are an important part of blood. When you cut or scrape your skin, you bleed because blood vessels have been opened. As soon as bleeding starts, platelets begin to clump together in the damaged area. They form a plug that helps reduce blood loss, as shown in **Figure 2.** Platelets also release chemicals that react with proteins in plasma. The reaction causes tiny fibers to form. The fibers help create a blood clot.

White Blood Cells

Sometimes *pathogens* (PATH uh juhnz)—bacteria, viruses, and other microscopic particles that can make you sick—enter your body. When they do, they often meet *white blood cells,* or WBCs. WBCs, shown in **Figure 3,** help keep you healthy by destroying pathogens. WBCs also help clean wounds.

WBCs fight pathogens in several ways. Some WBCs squeeze out of blood vessels and move around in tissues, searching for pathogens. When they find a pathogen, they destroy it. Other WBCs release antibodies. *Antibodies* are chemicals that identify or destroy pathogens. WBCs also keep you healthy by destroying body cells that have died or been damaged. Most WBCs are made in bone marrow. Some WBCs mature in the lymphatic system.

Reading Check **Why are WBCs important to your health?**



Figure 3 *White blood cells defend the body against pathogens. These white blood cells have been colored yellow to make their shape easier to see.*

Blood

Blood is part of the cardiovascular system. It travels through miles and miles of blood vessels to reach every cell in your body. So, you must have a lot of blood, right?

Well, actually, an adult human body has about 5 L of blood. Your body probably has a little less than that. All the blood in your body would not fill two 3 L soda bottles.

Learn

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od pressure is

ABO blood types
y are important.

What Is Blood?

Your cardiovascular system is made up of your heart, your blood vessels, and blood. **Blood** is a connective tissue made up of plasma, red blood cells, platelets, and white blood cells. Blood carries oxygen and nutrients to all parts of your body.

Reading Check **What are the four main components of blood?** *(See the Appendix for answers to Reading Checks.)*

STRATEGY

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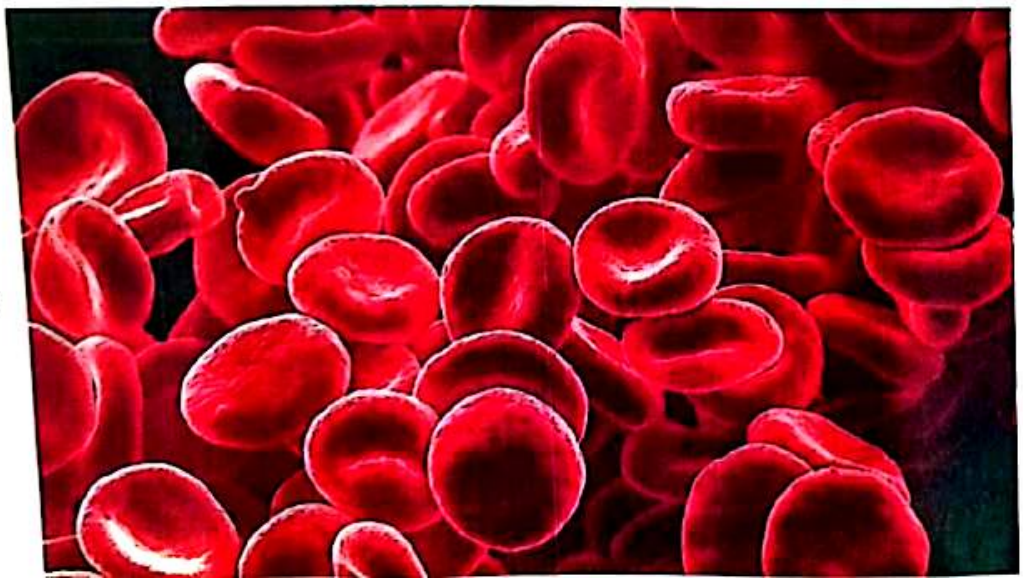
Plasma

The fluid part of the blood is called plasma (PLAZ muh). *Plasma* is a mixture of water, minerals, nutrients, sugars, proteins, and other substances. Red blood cells, white blood cells, and platelets are found in plasma.

uid that carries gases,
wastes through the
is made up of plasma,
s, platelets, and white

Red Blood Cells

Most blood cells are *red blood cells*, or RBCs. RBCs, such as the ones shown in **Figure 1**, take oxygen to every cell in your body. Cells need oxygen to carry out their functions. Each RBC has hemoglobin (HEE moh GLOH bin). *Hemoglobin* is an oxygen-carrying protein. Hemoglobin clings to the oxygen you inhale. RBCs can then transport oxygen throughout the body. Hemoglobin also gives RBCs their red color.



Red blood
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Two Types of Circulation

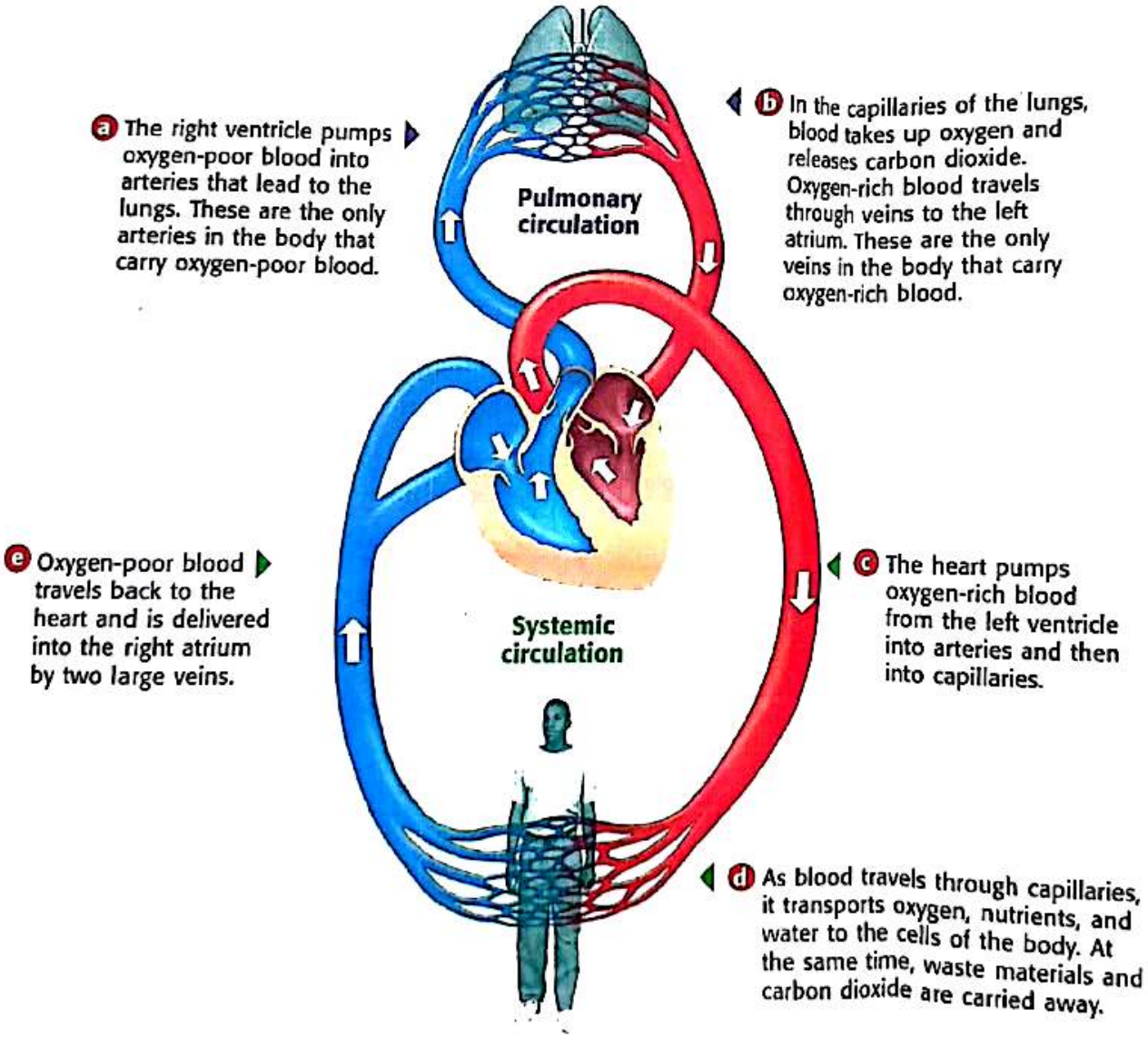
Where does blood get the oxygen to deliver to your body? From your lungs! Your heart pumps blood to the lungs. In the lungs, carbon dioxide leaves the blood and oxygen enters the blood. The oxygen-rich blood then flows back to the heart. This circulation of blood between your heart and lungs is called **pulmonary circulation** (PUL muh NER ee SUHR kyoo LAY shuhn).

The oxygen-rich blood returning to the heart from the lungs is then pumped to the rest of the body. The circulation of blood between the heart and the rest of the body is called **systemic circulation** (sis TEM ik SUHR kyoo LAY shuhn). Both types of circulation are shown in **Figure 5.**

pulmonary circulation the flow of blood from the heart to the lungs and back to the heart through the pulmonary arteries, capillaries, and veins

systemic circulation the flow of blood from the heart to all parts of the body and back to the heart

Figure 5 The Flow of Blood Through the Body



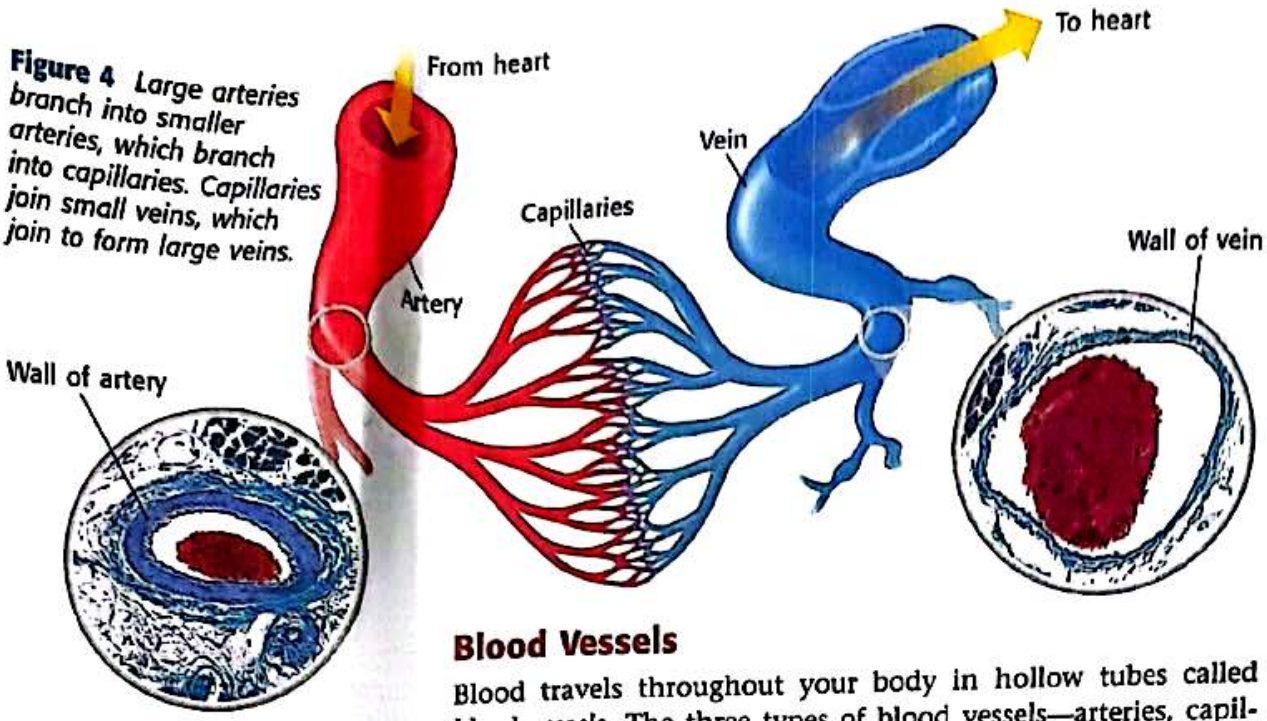


Figure 4 *Large arteries branch into smaller arteries, which branch into capillaries. Capillaries join small veins, which join to form large veins.*

Blood Vessels

Blood travels throughout your body in hollow tubes called *blood vessels*. The three types of blood vessels—arteries, capillaries, and veins—are shown in **Figure 4.**

artery a blood vessel that carries blood away from the heart to the body's organs

capillary a tiny blood vessel that allows an exchange between blood and cells in other tissue

vein in biology, a vessel that carries blood to the heart

Arteries

A blood vessel that carries blood away from the heart is an **artery.** Arteries have thick walls, which contain a layer of smooth muscle. Each heartbeat pumps blood into your arteries at high pressure. This pressure is your *blood pressure.* Artery walls stretch and are usually strong enough to stand the pressure. Your *pulse* is caused by the rhythmic change in your blood pressure.

Capillaries

Nutrients, oxygen, and other substances must leave blood and get to your body's cells. Carbon dioxide and other wastes leave body cells and are carried away by blood. A **capillary** is a tiny blood vessel that allows these exchanges between body cells and blood. These exchanges can take place because capillary walls are only one cell thick. Capillaries are so narrow that blood cells must pass through them in single file. No cell in the body is more than three or four cells away from a capillary.

Veins

After leaving capillaries, blood enters veins. A **vein** is a blood vessel that carries blood back to the heart. As blood travels through veins, valves in the veins keep the blood from flowing backward. When skeletal muscles contract, they squeeze nearby veins and help push blood toward the heart.

Reading Check **Describe the three types of blood vessels.**

The Heart

Your *heart* is an organ made mostly of cardiac muscle tissue. It is about the size of your fist and is almost in the center of your chest cavity. Like hearts of all mammals, your heart has a left side and a right side that are separated by a thick wall. The right side of the heart pumps oxygen-poor blood to the lungs. The left side pumps oxygen-rich blood to the body. As you can see in **Figure 2,** each side has an upper chamber and a lower chamber. Each upper chamber is called an *atrium* (plural, *atria*). Each lower chamber is called a *ventricle*.

Flaplike structures called *valves* are located between the atria and ventricles and in places where large arteries are attached to the heart. As blood moves through the heart, these valves close to prevent blood from going backward. The "lub-dub, lub-dub" sound of a beating heart is caused by the valves closing. **Figure 3** shows the flow of blood through the heart.

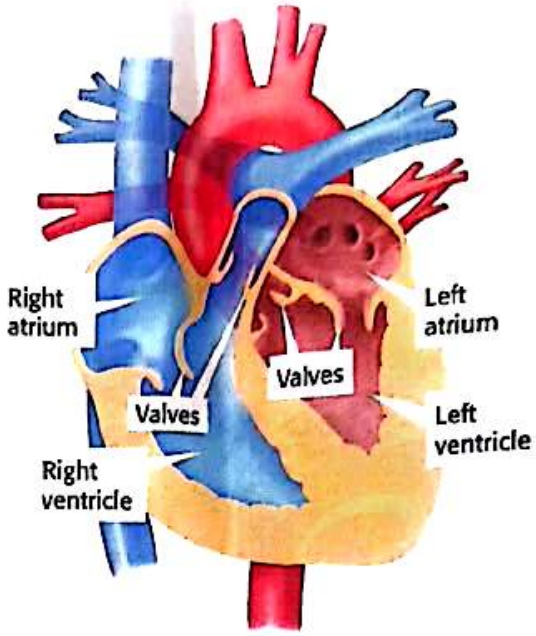
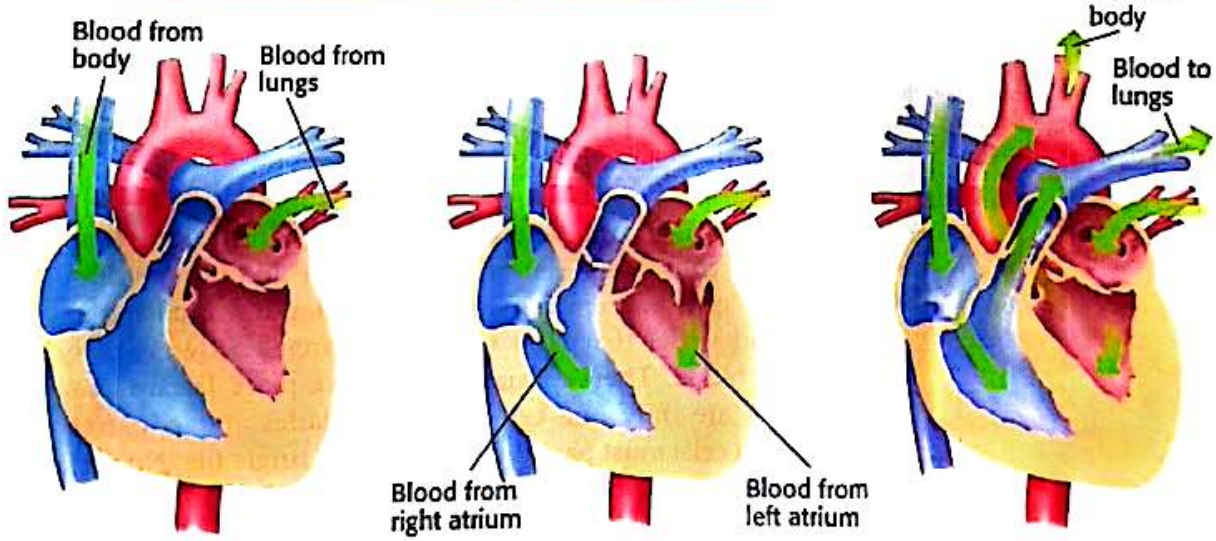


Figure 2 *The heart pumps blood through blood vessels. The vessels carrying oxygen-rich blood are shown in red. The vessels carrying oxygen-poor blood are shown in blue.*

Figure 3 The Flow of Blood Through the Heart



1. Blood enters the atria first. The left atrium receives oxygen-rich blood from the lungs. The right atrium receives oxygen-poor blood from the body.
2. When the atria contract, blood is squeezed into the ventricles.
3. While the atria relax, the ventricles contract and push blood out of the heart. Blood from the right ventricle goes to the lungs. Blood from the left ventricle goes to the rest of the body.

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SECTION 1

The Cardiovascular System

When you hear the word **heart,** what do you think of first? Many people think of romance. Some people think of courage. But the heart is much more than a symbol of love or bravery. Your heart is an amazing pump.

The heart is an organ that is part of your cardiovascular system. The word *cardio* means "heart," and the word *vascular* means "blood vessel." The blood vessels-arteries, capillaries, and veins-carry blood pumped by the heart.

What You Will Learn

- List five main parts of the cardiovascular system, and describe their functions.
- Describe the two types of circulation of blood in the body.
- List four cardiovascular problems.

Vocabulary

- cardiovascular system
- artery
- capillary
- vein
- pulmonary circulation
- systemic circulation

READING STRATEGY

Paired Summarizing Read this section silently. In pairs, take turns summarizing the material. Stop to discuss ideas that seem confusing.

cardiovascular system a collection of organs that transport blood throughout the body

Your Cardiovascular System

Your heart, blood, and blood vessels make up your **cardiovascular system** (KAR dee OH VAS kyoo luhr SIS tuhm). Your heart creates pressure when it beats. This pressure moves blood throughout your body. **Figure 1** shows your heart, major arteries, and major veins.

The cardiovascular system helps maintain homeostasis by performing many functions. For example, this system helps maintain your body by carrying nutrients to your cells and by removing wastes from your cells. This system also helps in regulation by carrying chemical signals called *hormones* throughout the body.

Reading Check What are the main parts of the cardiovascular system? *(See the Appendix for answers to Reading Checks.)*

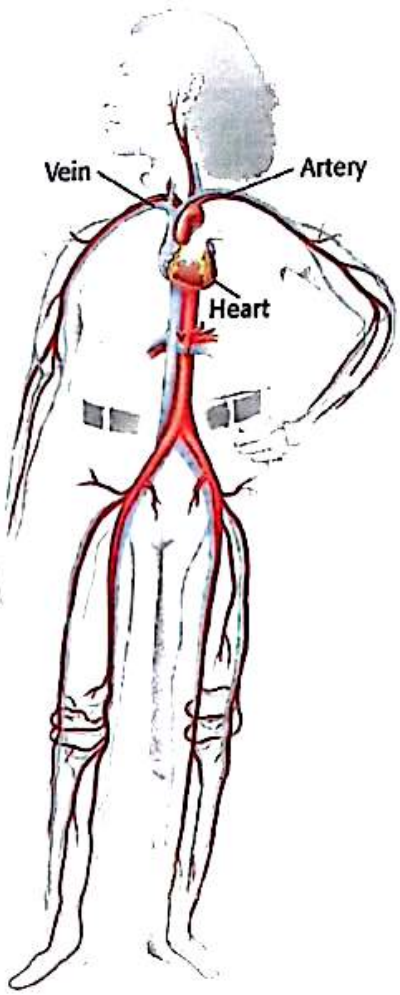


Figure 1 *The cardiovascular system carries blood to every cell in your body.*

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More Endocrine Glands

Your body has several other endocrine glands. Some of these glands have many functions. For example, your pituitary gland stimulates skeletal growth and helps the thyroid gland work properly. It also regulates the amount of water in the blood. And the pituitary gland stimulates the birth process in women.

Your thyroid gland is very important during infancy and childhood. Thyroid hormones control the secretion of growth hormones for normal body growth. Thyroid hormones also control the development of the central nervous system. And they control your metabolism. *Metabolism* is the sum of all the chemical processes that take place in an organism.

Your thymus gland is important to your immune system. Cells called *killer T cells* grow and mature in the thymus gland. These T cells help destroy or neutralize cells or substances that invade your body. The names and some of the functions of endocrine glands are shown in **Figure 2.**

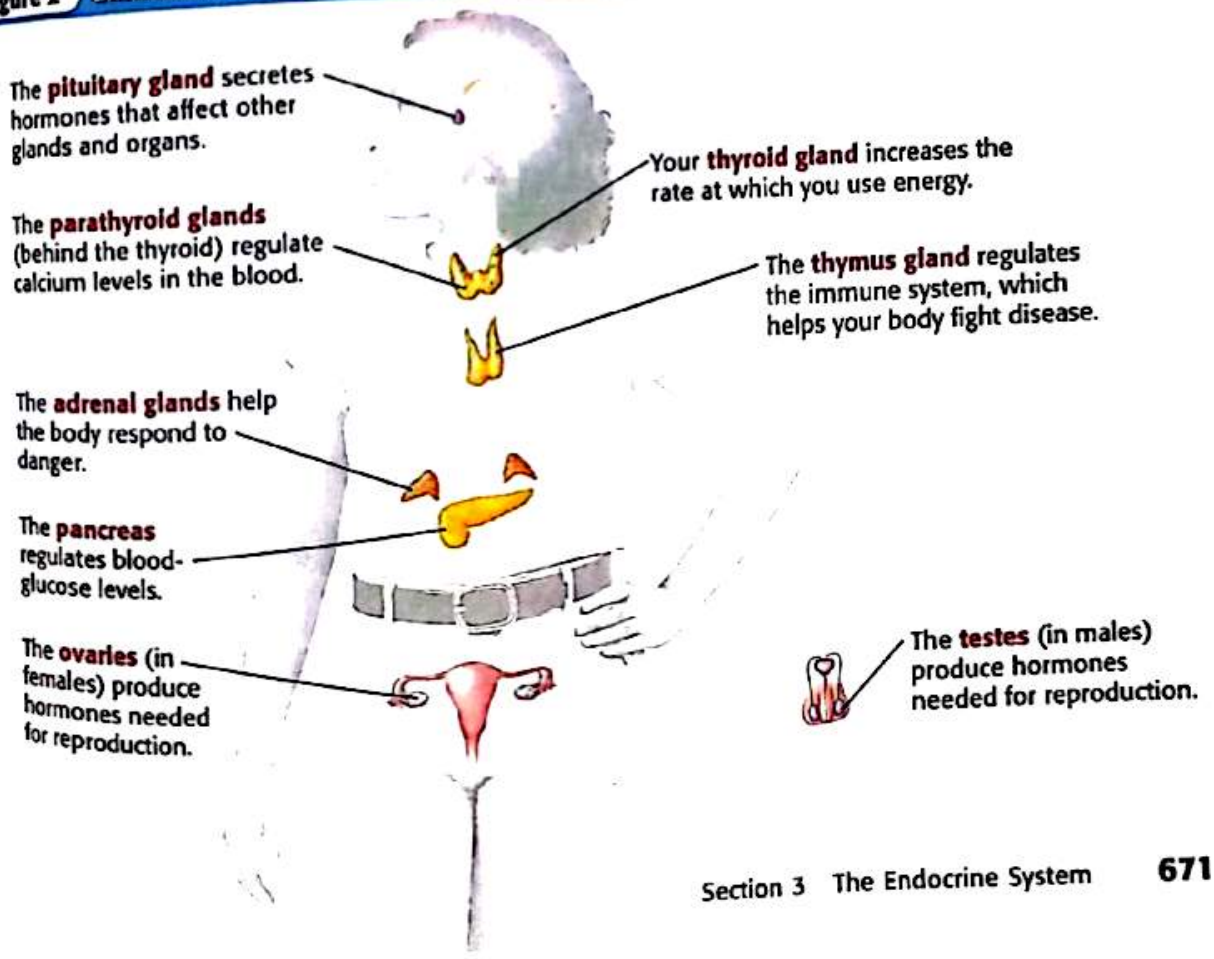
Reading Check **Name two endocrine glands, and explain why they are important to your body.** *(See the Appendix for answers to Reading Checks.)*

hormone a substance that is made in one cell or tissue and that causes a change in another cell or tissue in a different part of the body

CONNECTION TO Language Arts

WRITING SKILL **Fight or Flight?** Write a paragraph describing a time when you had a fight-or-flight experience. Include in your description the following terms: *hormones, fight-or-flight,* and *epinephrine.* If you cannot think of a personal experience, write a short story describing someone else's fight-or-flight experience.

Figure 2 Endocrine Glands and Their Functions



SECTION 3

The Endocrine System

Have you ever heard of an epinephrine (EP uh NEPH rin) rush? You might have had one without realizing it. Exciting situations, such as riding a roller coaster or watching a scary movie, can cause your body to release epinephrine.

Epinephrine is one of the body's chemical messengers made by the endocrine system. Your endocrine system regulates body processes, such as fluid balance, growth, and development.

What You Will Learn

- Explain why the endocrine system is important to the body.
- Identify five glands of the endocrine system, and describe what their hormones do.
- Describe how feedback mechanisms stop and start hormone release.
- Name two hormone imbalances.

Vocabulary

endocrine system
gland
hormone

READING STRATEGY

Discussion Read this section silently. Write down questions that you have about this section. Discuss your questions in a small group.

Hormones as Chemical Messengers

The **endocrine system** controls body functions by using chemicals that are made by the endocrine glands. A **gland** is a group of cells that make special chemicals for your body. Chemical messengers made by the endocrine glands are called hormones. A **hormone** is a chemical messenger made in one cell or tissue that causes a change in another cell or tissue in another part of the body. Hormones flow through the bloodstream to all parts of the body. Thus, an endocrine gland near your brain can control an organ that is somewhere else in your body.

Endocrine glands may affect many organs at one time. For example, in the situation shown in **Figure 1**, the adrenal glands release the hormone *epinephrine*, which is sometimes called *adrenaline*. Epinephrine increases your heartbeat and breathing rate. This response is called the "fight-or-flight" response. When you are frightened, angry, or excited, the "fight-or-flight" response prepares you to fight the danger or to run from it.

endocrine system a collection of glands and groups of cells that secrete hormones that regulate growth, development, and homeostasis

gland a group of cells that make special chemicals for the body



Figure 1 *When you have to move quickly to avoid danger, your adrenal glands make more blood glucose available for energy.*

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Muscle Injury

Any exercise program should be started slowly. Starting slowly means you are less likely to get hurt. You should also warm up for exercise. A *strain* is an injury in which a muscle or tendon is overstretched or torn. Strains often happen because a muscle has not been warmed up. Strains also happen when muscles are worked too hard.

People who exercise too much can hurt their tendons. The body can't repair an injured tendon before the next exercise session. So, the tendon becomes inflamed. This condition is called *tendinitis*. Often, a long rest is needed for the injured tendon to heal.

Some people try to make their muscles stronger by taking drugs. These drugs are called *anabolic steroids* (A nuh BAH lik STER OIDZ). They can cause long-term health problems. Anabolic steroids can damage the heart, liver, and kidneys. They can also cause high blood pressure. If taken before the skeleton is mature, anabolic steroids can cause bones to stop growing.

Reading Check What are the risks of using anabolic steroids?

MATH PRACTICE

Runner's Time

Jan has decided to enter a 5 km road race. She now runs 5 km in 30 min. She would like to decrease her time by 15% before the race. What will her time be when she reaches her goal?

SECTION Review

Summary

- The three kinds of muscle tissue are smooth muscle, cardiac muscle, and skeletal muscle.
- Skeletal muscles work in pairs. Skeletal muscles contract to move bones.
- Resistance exercise improves muscle strength. Aerobic exercise improves heart strength and muscle endurance.
- Strains are injuries that affect muscles and ten-

Using Key Terms

1. In your own words, write a definition for the term *muscular system*.

Understanding Key Ideas

2. Muscles
   a. work in pairs.
   b. move bones by relaxing.
   c. get smaller when exercised.
   d. All of the above
3. Describe three kinds of muscle.
4. List two kinds of exercise. Give an example of each.
5. Describe two muscular system injuries.

Math Skills

6. If Trey can do one curl-up every 2.5 s, about how long will it take

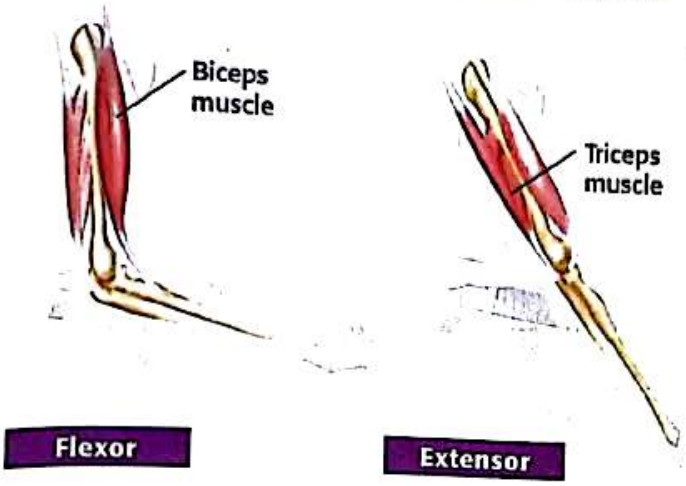
Critical Thinking

7. **Applying Concepts** Describe some of the muscle action needed to pick up a book. Include flexors and extensors in your description.
8. **Predicting Consequences** If aerobic exercise improves heart strength, what likely happens to heart rate as the heart gets stronger? Explain your answer.



Figure 2 A Pair of Muscles in the Arm

Skeletal muscles, such as the biceps and triceps muscles, work in pairs. When the biceps muscle contracts, the arm bends. When the triceps muscle contracts, the arm straightens.



muscular system the organ system whose primary function is movement and flexibility

Movement

Skeletal muscles can make hundreds of movements. You can see many of these movements by watching a dancer, a swimmer, or even someone smiling or frowning. When you want to move, signals travel from your brain to your skeletal muscle cells. The muscle cells then contract, or get shorter.

Muscles Attach to Bones

Strands of tough connective tissue connect your skeletal muscles to your bones. These strands are called *tendons*. When a muscle that connects two bones gets shorter, the bones are pulled closer to each other. For example, tendons attach the biceps muscle to a bone in your shoulder and to a bone in your forearm. When the biceps muscle contracts, your forearm bends toward your shoulder.

Muscles Work in Pairs

Your skeletal muscles often work in pairs. Usually, one muscle in the pair bends part of the body. The other muscle straightens part of the body. A muscle that bends part of your body is called a *flexor* (FLEKS uhr). A muscle that straightens part of your body is an *extensor* (ek STEN suhr). As shown in **Figure 2**, the biceps muscle of the arm is a flexor. The triceps muscle of the arm is an extensor.

Reading Check Describe how muscles work in pairs. *(See the Appendix for answers to Reading Checks.)*

SCHOOL to HOME

Power in Pairs

Ask a parent or guardian to sit in a chair and place a hand palm up under the edge of a table. Tell your parent to apply gentle upward pressure. Feel the front and back of your parent's upper arm. Next, ask your parent to push down on top of the table. Feel your parent's arm again. What did you notice about the muscles in your parent's arm when he or she was pressing up? pushing down?

ACTIVITY

SECTION 3

The Muscular System

Have you ever tried to sit still, without moving any muscles at all, for one minute? It's impossible! Somewhere in your body, muscles are always working.

What You Will Learn

- List three kinds of muscle tissue.
- Describe how skeletal muscles move bones.
- Compare aerobic exercise with resistance exercise.
- Describe two muscular system injuries.

Vocabulary

muscular system

READING STRATEGY

Discussion Read this section silently. Write down questions that you have about this section. Discuss your questions in a small group.

Your heart is a muscle. Muscles make you breathe. And muscles hold you upright. If all of your muscles rested at the same time, you would collapse. The **muscular system** is made up of the muscles that let you move.

Kinds of Muscle

Figure 1 shows the three kinds of muscle in your body. *Smooth muscle* is found in the digestive tract and in the walls of blood vessels. *Cardiac muscle* is found only in your heart. *Skeletal muscle* is attached to your bones for movement. Skeletal muscle also helps protect your inner organs.

Muscle action can be voluntary or involuntary. Muscle action that is under your control is *voluntary*. Muscle action that is not under your control is *involuntary*. Smooth muscle and cardiac muscle are involuntary muscles. Skeletal muscles can be both voluntary and involuntary muscles. For example, you can blink your eyes anytime you want to. But your eyes will also blink automatically.

Figure 1 Three Kinds of Muscle



Skeletal muscle enables bones to move.

Smooth muscle moves food through the digestive system.

Cardiac muscle pumps blood around the body.